

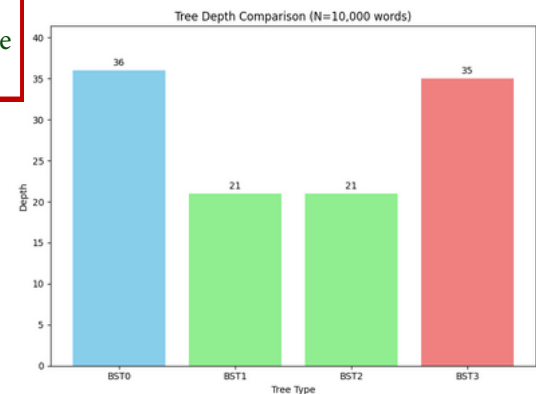
# Optimized Word Search Using BST Variants

## Tree Construction Characteristics And Word Search Efficiency (N=10,000 words)

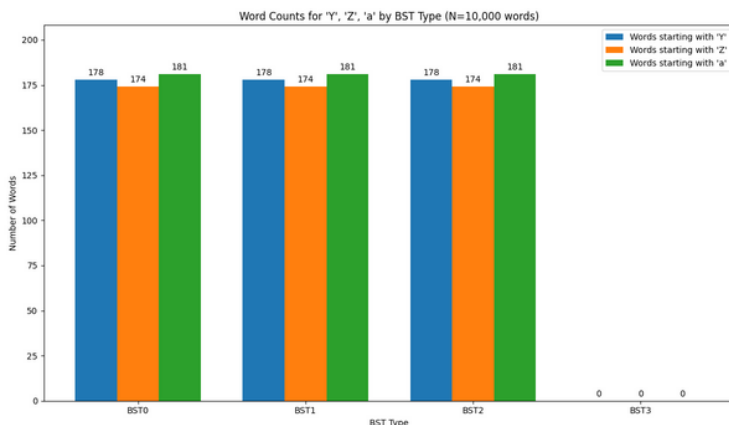
**Important Note:** The tree traversal and manipulation algorithms, including node insertion, deletion and rotations are implemented recursively (using the systems's stack)

**Observation (Depths):** Special trees BST1 & BST2 (depth 21) are shorter than BST0 (36) & BST3 (35), which is good for speed (see Figure 1).

**Figure 1: Depth Comparison**

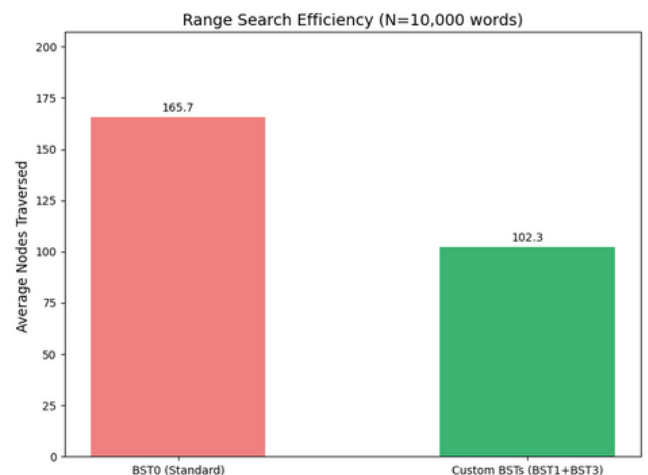


**Figure 2: Word Counts for 'Y', 'Z', 'a' by BST Type**



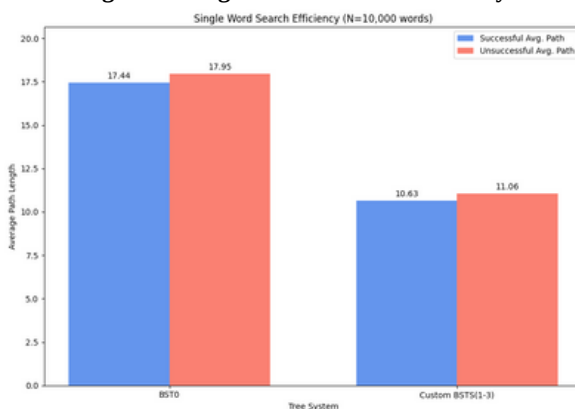
**Observation (YZa Counts):** Words starting with 'Y', 'Z', or 'a' correctly went into BST1 & BST2, and not into BST3, as planned (see Figure 2).

**Figure 3: BST Range Search Efficiency**



**Observation :** For range searches, Custom BSTs checked fewer nodes (102.3) than BST0 (165.7), making them faster (see Figure 3).

**Figure 4: Single Word Search Efficiency**



**Conclusion (Search Efficiency):** Custom BSTs find words much faster than BST0. This is true when words are found (average path 10.63 vs 17.44) and when words are not found (average path 11.06 vs 17.95) (see Figure 4).

## Overall Conclusion

Using 10,000 words, the Custom BSTs system is better than a normal BST0 for finding single words. The special trees (BST1, BST2) are shorter, leading to quicker searches. The way words were stored in these trees was correct. Custom BSTs are clearly faster.